

2.2 Add and Subtract Rational Expressions

DO IT NOW!

$$LCD = 30$$

$$\text{a) } \cancel{x}^1 \frac{1}{\cancel{x}^6} + \frac{1 \times 6}{5 \times 6}$$

$$= \frac{5}{30} + \frac{6}{30}$$

$$= \frac{11}{30}$$

Note: the product of the denominators will give a common denominator (but not always the lowest common denominator)

b) Simplify and state restrictions

$$\frac{x^2 - 1}{x^2 - 4} \times \frac{x^2 + 3x - 4}{x^2 + 5x + 4} = \frac{(x-1)\cancel{(x+1)}}{(x-2)(x+2)} \times \frac{\cancel{(x+4)}(x-1)}{\cancel{(x+1)}\cancel{(x+1)}}$$

$$= \frac{(x-1)^2}{(x-2)(x+2)} ; x \neq 2, -2, -4, -1$$

c) $\frac{2x}{3y} + \frac{y}{6x}$ $LCD = 6xy$

$$= \frac{4x^2}{6xy} + \frac{y^2}{6xy} \quad ; \begin{matrix} x \neq 0 \\ y \neq 0 \end{matrix}$$

$$= \frac{4x^2 + y^2}{6xy} \quad ; \begin{matrix} x \neq 0 \\ y \neq 0 \end{matrix}$$

Add and Subtract Rational Expressions With Monomial Denominators

a) $\frac{1}{5x} + \frac{1}{2x}$ $LCD = 10x$

$$\frac{2}{10x} + \frac{5}{10x}$$

$$= \frac{2}{10x} + \frac{5}{10x} \quad ; x \neq 0$$

$$= \frac{7}{10x} \quad ; x \neq 0$$

1. factor denominators if possible
2. get a common denominator
3. re-write expression with a common denominator
4. add/subtract the numerator
(keep denominator the same)
5. simplify where possible
6. state restrictions (throughout process)

b)

$$\frac{ab^2 + 2}{2ab^2} - \frac{b + 2}{2b}$$

$LCD = 2ab^2$

$$= \frac{ab^2 + 2}{2ab^2} - \frac{ab(b+2)}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{ab^2 + 2}{2ab^2} - \frac{ab^2 + 2ab}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{ab^2 + 2 - ab^2 - 2ab}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{2 - 2ab}{2ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{\cancel{2}(1-ab)}{\cancel{2}ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

$$= \frac{1-ab}{ab^2} \quad ; \begin{matrix} a \neq 0 \\ b \neq 0 \end{matrix}$$

Add and Subtract Rational Expressions with Polynomial Denominators

a)

$LCD: (x-3)(x+2)$

1. factor denominators if possible
2. get a common denominator
3. re-write expression with a common denominator
4. add/subtract the numerator (keep denominator the same)
5. simplify where possible
6. state restrictions (throughout process)

$$\frac{x+5}{x-3} + \frac{x-7}{x+2}$$

$$= \frac{(x+2)(x+5)}{(x+2)(x-3)} + \frac{(x-7)(x-3)}{(x+2)(x-3)} \quad ; x \neq -2, 3$$

$$= \frac{x^2 + 7x + 10}{(x+2)(x-3)} + \frac{x^2 - 10x + 21}{(x+2)(x-3)} \quad ; x \neq -2, 3$$

$$= \frac{2x^2 - 3x + 31}{(x+2)(x-3)} \quad ; x \neq -2, 3$$

b)

$$\frac{x+9}{x^2+2x-48} - \frac{x-9}{x^2-x-30}$$

$$LCD = (x-6)(x+5)(x+8)$$

$$= \frac{\overset{(x+5)}{\cancel{(x+5)}} \overset{(x+8)}{\cancel{(x+8)}} (x+9)}{\overset{(x+5)}{\cancel{(x+5)}} \overset{(x+8)}{\cancel{(x+8)}} (x-6)} - \frac{(x-9) \overset{(x+8)}{\cancel{(x+8)}} \overset{(x+5)}{\cancel{(x+5)}}}{(x-6) \overset{(x+8)}{\cancel{(x+8)}} \overset{(x+5)}{\cancel{(x+5)}}}$$

$$= \frac{(x+5)(x+9)}{(x+5)(x+8)(x-6)} - \frac{(x-9)(x+8)}{(x+5)(x+8)(x-6)}$$

$$= \frac{x^2+14x+45 - (x^2-1x-72)}{(x+5)(x+8)(x-6)}$$

$$= \frac{15x+117}{(x+5)(x+8)(x-6)} ; x \neq -5, -8, 6$$

c)

$$\frac{x-2}{x+2} + \frac{x+10}{x^2+6x+8}$$

$$LCD = (x+2)(x+4)$$

$$= \frac{\overset{(x+4)}{\cancel{(x+4)}} \overset{(x+2)}{\cancel{(x+2)}} (x-2)}{\overset{(x+4)}{\cancel{(x+4)}} \overset{(x+2)}{\cancel{(x+2)}}} + \frac{(x+10)}{(x+2)(x+4)}$$

$$= \frac{(x+4)(x-2)}{(x+4)(x+2)} + \frac{x+10}{(x+4)(x+2)}$$

$$= \frac{x^2+2x-8+x+10}{(x+4)(x+2)}$$

$$= \frac{x^2+3x+2}{(x+4)(x+2)} \leftarrow \text{factor}$$

$$= \frac{\cancel{(x+2)}(x+1)}{(x+4)\cancel{(x+2)}}$$

$$= \frac{x+1}{x+4} ; x \neq -4, -2$$

d)

$$\frac{2x}{x-1} - \frac{x+2}{x^2+3x-4}$$

$$LD = (x+4)(x-1)$$

$$= \frac{(x+4)2x}{(x+4)(x-1)} - \frac{x+2}{(x+4)(x-1)}$$

$$= \frac{2x(x+4)}{(x+4)(x-1)} - \frac{x+2}{(x+4)(x-1)}$$

$$= \frac{2x^2+8x-x-2}{(x+4)(x-1)}$$

$$= \frac{2x^2+7x-2}{(x+4)(x-1)} ; x \neq -4, 1$$

e)

$$\frac{a+1}{5-2a} - \frac{a-4}{2a-5}$$

Binomial expressions can differ by a factor of -1. Factor -1 from one of the denominators to identify the common denominator. Then simplify each expression and state the restrictions.

$$= \frac{a+1}{-1(-5+2a)} - \frac{a-4}{2a-5}$$

$$= \frac{(-1)a+1}{(-1)(-5+2a)} - \frac{a-4}{2a-5}$$

$$= \frac{-1(a+1)}{2a-5} - \frac{a-4}{2a-5}$$

$$= \frac{-1(a+1) - (a-4)}{2a-5}$$

$$= \frac{-a-1-a+4}{2a-5}$$

$$= \frac{-2a+3}{2a-5} ; a \neq \frac{5}{2}$$